

Changes from `paper3_main_v8` to `v11`

Consolidated record for the co-author. Covers three revision steps:

step	date	nature
v8 → v9	16 Sep 2026	every number regenerated from a single consistent pipeline
v9 → v10	17 Sep 2026	figures consolidated into panels; per-text figures moved to an appendix
v10 → v11	18 Sep 2026	statistics (permutation tests, Spearman), notation collisions, text corrections

Every number is produced by the released code from the tokenised texts with fixed seeds, and is reproduced by `scripts/paper_numbers.py data/corpus`, which writes `results/paper_numbers.md`.

Status column: `done` = already in the `.tex`; `pending` = agreed but not yet applied. Nothing marked `pending` should be taken as final.

0. What matters most, in four points

- The pipeline is reproducible.** The four Pearson correlations regenerate exactly (0.702, 0.295, 0.511, 0.571), verified against `scipy` by an assertion inside `paper_numbers.py`. Everything else in this document rests on that.
- The FGN surrogate is a permutation of the original text.** The rank-to-word mapping reproduces the empirical rank histogram exactly, so all three sequences — original, word-shuffled, FGN — share the same token multiset and therefore the same embedding covariance, PCA eigensystem and projected marginals, *exactly*. They differ only in the ordering. This is stronger than what the paper currently claims, and it exposed three stale passages that survived from v8 (§3.6).
- The significance tests changed, and two marks moved.** The parametric `-test` on assumes bivariate normality; the eigenvalue spectrum is strongly skewed, so it was replaced by a permutation test over the pairing of the two spectra. *War and Peace* drops from `***` to `**`, *Ulysses* from `**` to `*`. The substantive conclusions are unchanged.
- Four factual errors were found, all traceable to v8.** The burstiest direction of *Origin of Species* (PC2 → PC6, §3.5); the OOV rate ("5-15%" → under 2%, §3.9); the number of shuffled realisations behind the ACF curve (10 → 1, §3.9); and the claim that the FGN and

original covariance structures differ by a few percent (they are identical, §3.6). Three of the four were contradicted by the paper’s own tables or captions.

1. v8 → v9 — regenerated numbers

The earlier version mixed results from three separate Colab batches, each of which recomputed PCA and the FGN surrogate independently. The pipeline now computes them once per text, so the burstiness, ACF and DFA tables share the same surrogate (same , same realisation).

Two consequences worth stating plainly to a referee:

- The FGN surrogate was previously built from a differently tokenised stream (regex `\w+`). DFA exponents in the corpus table shift by up to 0.03 (*Ulysses*: 0.794 → 0.763).
- In the previous ACF batch the FGN of every text had been tuned to by a coding slip. The FGN autocorrelation is negligible either way, so the ACF conclusions are unchanged.

1.1 Corpus (status: done)

Alice in Wonderland removed (7,237 filtered tokens — too short for a reliable IET estimate at). Corpus is now 16 texts.

quantity	v8	v9+
texts	17	16
raw token range	27,000–565,000	56,000–565,000
filtered token range	12,000–321,000	17,000–236,000
DFA exponent range		

A column (FGN generator exponent) was added to the corpus table. is determined only to by the bisection tolerance and is therefore reported with two decimals throughout.

1.2 Four showcase texts (status: done)

text	v8 → v9	v8 → v9	v8 → v9
War and Peace	0.719 → 0.721	0.770 → 0.76	0.702 → 0.702
Origin of Species	0.769 → 0.771	0.804 → 0.80	0.294 → 0.295
Great Expectations	0.688 → 0.689	0.735 → 0.74	0.522 → 0.511

text	v8 → v9	v8 → v9	v8 → v9
Ulysses	0.794 → 0.763	0.839 → 0.80	0.578 → 0.571
Mean burstiness over the first 20 components at was previously reported without stating the quantile and with values from a different batch: 1.18 / 1.10 / 1.07 / 1.13 → 1.36 (War and Peace), 1.35 (Origin), 1.23 (Ulysses), 1.18 (Great Expectations), against – for both null models in every case.			

1.3 Corpus-level burstiness (status: done)

quantity	v8	v9+
	, mean 1.22	, mean 1.26
	99.4% of pairs	100% of pairs
	95.9%	95.6%
per-text mean ratio orig/surr		
FGN/surr ratio	, ,	, ,

The last row is the most useful change for the argument. In v8 the two null models were *significantly different* from each other, which was awkward to explain away as estimator noise. With the consistent pipeline they are statistically indistinguishable — the cleaner and stronger statement.

The abstract changed accordingly: "exceeds the FGN surrogate in of pairs" → "in all pairs".

The excess of the original over the nulls is now described as a mean excess (was: a consistent excess). The per-text spread is reported explicitly: for fifteen texts with *Principia Mathematica* an outlier at 1.73, sd .

1.4 ACF and DFA (status: done)

quantity	v8	v9+
GE: original range		

quantity	v8	v9+
GE: i.i.d. band $\sqrt{\quad}$		
GE: FGN lag-1	, 19/20 inside band	, 20/20 inside band
corpus		
corpus		
FGN pairs significant at lag 117.2%		10.9%
-test FGN vs 0		
GE:		
GE:		
GE: range		

A remark about PC1 lying outside the i.i.d. band ($\rho_1^{\text{FGN}}(1) = 0.023$) was **deleted**: with the consistent pipeline the value is 0.004, inside the band.

The residual FGN excess at lag 1 is now quantified rather than asserted: the significant cases concentrate on PC1 (56% of PC1 pairs against 8% for $\rho_1^{\text{FGN}}(1)$), consistent with the known GloVe artefact from rare transliterated proper nouns.

2. v9 → v10 — figure consolidation (status: done)

No numbers changed in this step. Structural only:

- Four per-text burstiness-spectrum figures → one 4-panel figure (`fig:burst_panel`), panels A–D).
- Four per-text quantile-sweep figures → one 4-panel figure (`fig:sweep_panel`).
- The four "semantic poles" figures were **removed**; the same information is in the PC tables (`tab:wrnpc_pc` and siblings), which are more precise and cost less space.
- Per-text eigenvalue spectra and IET distributions moved to a new appendix, *Additional per-text figures* (`app:per-text`), keeping only *War and Peace* in the main text as the worked example.
- Figure paths now follow the corpus filenames (`figures/<stem>/<stem>_NN_<name>.png`), so every figure is regenerable by `scripts/run_text.py`.
- Dropbox references replaced by a pointer to the code repository.

Net effect: roughly fifteen figures down to the panels plus the appendix.

3. v10 → v11 — statistics, notation, corrections

3.1 Notation: `p` was overloaded (status: partly done)

`p` meant both the **event rate** (Sec. 2.3, and in the robustness section at) and the **p-value** of the correlation and χ^2 -tests — both small decimals, a few pages apart. The event rate is structural and appears throughout, so the p-value was renamed `P`.

Seven occurrences converted (status: done): the four pairs in the per-text sections, and three χ^2 -test p-values in the corpus sections.

Four occurrences still open (status: **pending**) — they use `<` and `>` rather than `=`, so the first pass missed them:

- table `tab:single_text_summary` footnote: `$^*p<0.05$`, `$^{**}p<0.01$`, `$^{***}p<0.001$`, `no`
`mark: $p>0.05$`
- `t = 8.7, \quad p < 10^{-6}` (eq. `ratio_corpus`)
- `$t = 5.05$, $p < 10^{-6}$` (corpus ACF)

3.2 Three further symbol collisions (status: pending)

symbol	conflicting meanings	resolution
	matching condition in exponent (≈ 65 uses) vs DFA	matching condition $\rightarrow$ (6 occurrences)
	Zipf exponent vs diffusion	diffusion exponent $\rightarrow$, tied explicitly to the DFA of the stationary series
	retained tokens vs FGN	surrogate $\rightarrow$ (1 occurrence)
	surrogate word sequence	

and "Hurst" appear nowhere else in the paper, so this introduces no new collision.

3.3 Significance by permutation, not by the parametric test (status: pending)

The pairs are not independent draws: is a deterministic decreasing sequence and both spectra derive from the same trajectory. The χ^2 -test on is therefore not a valid significance test here. It is replaced by a permutation test over the pairing between the two spectra (pairings, seed 0), implemented in `scripts/pearsonr_spearman.py` and called from `paper_numbers.py`.

text	parametric	permutation	mark
War and Peace	0.702 0.0006	0.0055	*** → **
Origin of Species	0.295 0.207	0.169	none → none
Great Expectations	0.511 0.021	0.0136	* → *
Ulysses	0.571 0.009	0.0406	** → *

The parametric test was anticonservative by up to an order of magnitude, as expected when the predictor is strongly skewed.

Note for the reader of the table: *Ulysses* has a larger than *Great Expectations* but a larger . This is not an error — the permutation null depends on the shape of each text's eigenvalue spectrum.

3.4 Spearman added (status: pending)

The eigenvalue spectrum decays steeply, so Pearson is dominated by the leading components. The rank correlation was computed as a robustness check and turns out to carry information of its own:

text			
War and Peace	0.702	0.850	
Origin of Species	0.295	0.460	0.042
Great Expectations	0.511	0.579	0.0085
Ulysses	0.571	0.817	

in all four texts: the association between semantic variance and burstiness is **monotone but not linear**, and Pearson understates it.

$P = 0.0000$ in the regenerated output is the floor of permutations () and is reported as , not as zero.

3.5 Text correction: the burstiness maximum of Darwin (status: pending)

The body text of the *Origin of Species* subsection states that PC2 is the burstiest direction. It is not: the maximum is at **PC6**, , against . The v10 caption of `fig:burst_panel` already gives the maxima correctly as PC3, PC6, PC3, PC2, so the paper currently contradicts itself. The body text is a leftover from v9, when the now-removed per-text figure `fig:darwin_burst` carried the PC2 claim.

This matters beyond the correction. The subsection used the weak correlation as its argument:

The weak eigenvalue–burstiness correlation, (), shows especially clearly that the direction of greatest semantic variance need not be the direction of greatest temporal clustering.

That argument no longer survives contact with Spearman: , is a significant association. The displacement of the maximum to PC6 says the same thing directly, needs no correlation measure at all, and holds in every text (PC3, PC6, PC3, PC2 — never PC1). The subsection is rewritten around that fact, with the two correlations reported as description rather than as the argument.

3.6 The FGN surrogate is a permutation — and three stale passages (status: pending)

Verified directly: `Counter(tokens_all)` of the FGN surrogate equals that of the original, for every showcase text, and the filtered counts agree to the token at , and . Multiset identity at the unfiltered level makes the result independent of by construction, not by observation.

text	ratio multiset identical		
War and Peace	235,933	235,933	1.0000 yes
Origin of Species	58,913	58,913	1.0000 yes
Great Expectations	64,410	64,410	1.0000 yes
Ulysses	101,489	101,489	1.0000 yes

Consequences. By Eq. `sigma_invariant`, $\Sigma_{\text{FGN}} = \Sigma_{\text{orig}}$ **exactly**: the original eigenvectors are the surrogate's own principal directions. The design can now be stated in one line — *the three sequences are permutations of one another and differ only in the ordering of visits through a fixed semantic geometry* — and the two null models differ in exactly one ingredient, the rank-level scaling.

Three passages assert the opposite and are all v8 leftovers, from when the FGN was tokenised with a different regex and the multisets genuinely differed:

- `rem:fgn_cov`: "does not impose covariance invariance as an algebraic constraint ... empirically very close" → the invariance is algebraic.
- §Text preprocessing: "increasing to 150 or 200 ... begins to degrade the alignment between the original and FGN covariance structures" → no such degradation is possible; sentence deleted.

- §FGN surrogate generation: "any residual difference is below ϵ in eigenvalue and below δ in eigenvector angle" $\rightarrow$ the difference is zero.

A fourth passage becomes redundant rather than wrong: the sentence added earlier today noting that ϵ need not equal δ is removed, since the two are equal by construction.

One further consequence, in the paper's favour. The code computes the event threshold once from the original series and applies it to all three (`th = thresholds(an.pca.proj, q)`), whereas §2.3 describes each series being thresholded at its own quantile. The two conventions coincide exactly, because permutations share a marginal distribution — so the event rate is exactly q for all three sequences, and this can now be stated as a consequence rather than assumed.

3.7 Does the frequency filter destroy the rank-level scaling? (status: new result)

A referee can reasonably ask whether the FGN's projected series show ϵ because rank-level memory does not reach embedding space — the paper's claim — or simply because the top- k filter destroyed that memory before the projection. The question was settled by measuring the DFA exponent of the *filtered* rank sequence for both the original and the surrogate.

text	unfiltered	(pipeline) original,	filtered FGN,	filtered
War and Peace	0.693	0.721	0.674	0.633
Origin of Species	0.726	0.771	0.694	0.645
Great Expectations	0.672	0.689	0.609	0.567
Ulysses	0.805	0.763	0.772	0.629

The filter does not destroy the scaling. Every filtered value stays well above ϵ ; the FGN arrives at the projection stage still carrying rank-level memory of ϵ — ϵ , and its projections nevertheless show none. The claim is supported, now with a number behind it.

But the filter degrades the surrogate more than the original, in every text, and markedly so for *Ulysses* (ϵ against ϵ). The matching ϵ is imposed before the filter, so at the level where the comparison is made the null carries somewhat less memory than intended. This is a handicap to the null model, not a licence for it, and is worth stating rather than leaving to be found.

Caveat on these four columns: they were computed with `projection_alpha`, which does not reproduce the pipeline's own `alpha_text` (discrepancies up to 0.045, and of opposite sign for *Ulysses*). The three columns are comparable with one another but should not be

printed alongside and until recomputed with the estimator used inside `fgn_surrogate`.

3.8 Burstiness beyond the retained (status: new result)

The paper retains components. Whether anything changes beyond that was previously asserted without evidence; it has now been measured by refitting the PCA with

text	cum. var.	20	50	100	mean	1-20	21-50 min	21-50
War and Peace	37.8%		57.1%	76.4%	1.358		1.138	1.085
Origin of Species	38.1%		59.0%	78.6%	1.350		1.188	1.095
Great Expectations	38.5%		59.0%	78.7%	1.176		1.112	1.041
Ulysses	35.0%		53.2%	72.7%	1.234		1.120	1.041

Every component up to stays above the independent-event baseline in every text; the worst case, 1.041, is still above it. Variance per component decays from over the first twenty to over components 21-50 and over 51-100, and mean burstiness falls by 36-61% of its excess over the baseline.

So the qualitative picture does hold, and the honest formulation states both halves: burstiness persists well beyond the retained range while decreasing in magnitude. is a choice about interpretability, not a claim that nothing happens further out.

The refit reproduces the stored projections to , which is the float32 precision floor of the GloVe vectors — not a sign flip, which would have produced an error of order unity and would have changed , since the threshold is applied to the upper tail only.

3.9 Errors in the pipeline description (status: pending)

Four statements in the methodology do not describe the code that produced the results.

- **OOV rate.** "The OOV rate after filtering is typically - for Victorian English texts" — the paper's own summary table reports 1.6%, 0.4%, 1.3% and 0.7%, confirmed independently from the pre- and post-lookup token counts (1.58 / 0.44 / 1.34 / 0.73%). Wrong by an order of magnitude, and contradicted fifteen pages later.
- **Number of shuffled realisations.** The code uses three different conventions — 20 realisations for the burstiness spectrum (`n_surr = 20`), a single realisation for the ACF (`a.pca.proj[a.perms[0]]`, so labelled in the source), and 10 for the DFA band (`N_SURR_DFA`). The methodology describes a single permutation throughout, and the ACF figure caption claims "mean over 10 realisations", which is wrong on both counts. The DFA caption is correct.

- **PCA dimensions.** " components for eigenvalue analysis" and describe nothing in the code: `fit_pca` retains all 300 eigenvalues but only eigenvectors, entries of `explained_variance_ratio`, and an projection matrix, with .
- in Eq. `fgn_dfa_match` appears exactly once in the paper and conflates the generator exponent with the measured exponent ; the pipeline section states the relationship correctly.

Also stale, inside commented-out source that the authors intend to keep: the shuffle block states against `n_surr = 20` in `config.py`, and contains a sentence missing its first half. Both should carry a `% STALE` marker so they are not restored unexamined.

3.10 Editorial changes (status: done)

- §1.2 retitled *The Altmann et al. mechanism* → *From word-level burstiness to topical organisation*. The mechanism is joint work and the old title over-attributed it; the citation remains in the first sentence.
- "biological variation" → "geographical distribution" in the list of example topics, so the examples match the semantic poles actually found (Darwin's PC2 is geographical).
- stated once, where the burstiness spectrum is defined. The accompanying claim about higher-order components was initially written without evidence; it has since been measured and restated with figures (§3.8), and is a candidate for the robustness section of the Results rather than the introduction.
- Redundant restatements of "PCA maximises variance, not burstiness" and of the word-shuffled null's properties compressed.
- §Logical role of the null models now opens with a table (`tab:null_taxonomy`) setting out which statistical ingredient each sequence retains, and closes with the joint conclusion that both answers are negative and non-redundant. Given §3.6, the embedding geometry row reads *exact* for both null models, so the two differ in exactly one ingredient.
- `rem:fgn_entropy` concerns rank proximity versus semantic proximity and mentions no entropy; the label should be renamed while it is cited in only one place.

4. Open items

Ordered by consequence, not by effort.

1. Rewrite `rem:fgn_cov`, delete the / degradation sentence and the "below 3% / below 5%" passage (§3.6). These upgrade three hedges into an exact statement.
2. Correct the OOV rate, the ACF caption's realisation count, the passage and (§3.9).
3. Rewrite the *Origin of Species* subsection around the PC6 maximum (§3.5), and check the other three per-text subsections for the same class of error.
4. Apply the four remaining `p` → `P` (§3.1) and the three symbol renames (§3.2).

5. Table `tab:single_text_summary`: new marks on , new column, caption naming the permutation test.
6. State the three shuffled-realisation conventions in the methodology, and the fixed seeds (`pca_seed=42`, `fgn_seed=42`, `n_surr=20` with seeds 0–19, permutations with seed 0).
7. Recompute the filtered-rank DFA of §3.7 with the estimator used inside `fgn_surrogate` before quoting those numbers in the paper.
8. Decide where the result belongs: the robustness section of the Results, beside the quantile sweep, is the natural home.
9. `paper_numbers.py` sections 2–5 still carry hardcoded "(old ...)" strings from v8. The `.tex` itself is correct and in sync; only the script's comparison column is stale, so its output for those sections currently flags differences that do not exist.

5. Reproducing everything



bash

```
python scripts/run_corpus.py data/corpus      # corpus_*.csv
python scripts/paper_numbers.py data/corpus   # results/paper_numbers.md
python scripts/run_text.py data/corpus/<text>  # per-text figures
```

`paper_numbers.py` asserts that the permutation implementation reproduces `scipy.stats.pearsonr` to on every text, so a silent divergence between the two code paths cannot go unnoticed.

Permutation parameters are fixed: pairings, seed 0. Both belong in the methodology section alongside the test itself.

Section 6 of `results/paper_numbers.md` now carries the verifications introduced above: against , multiset identity at three values of , the filtered-rank DFA exponents, and the burstiness refit. Each carries its own guard — a calibration column for the DFA estimator, an assertion that the refit reproduces the stored projections, and an assertion on the number of components returned, so that a silent truncation cannot be mistaken for a result.